

Capstone Design

Project: Verified Study Buddy

1. Capstone Design

Teach the learning cycle

Primary target: Explain the mechanism and terminology of capstone requirements and corpus

Session · Sequence and minutes · Resources

Session: Session 1 · Understand · 45 min

Sequence and minutes: 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.

Resources: Sections 1–6; record the brief recall in the workbook.

Session: Session 2 · Practise · 45 min

Sequence and minutes: 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.

Resources: Sections 7–10; use the matching workbook W2 and misconception check.

Session: Session 3 · Investigate and transfer · 45 min

Sequence and minutes: 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.

Resources: Sections 9 and 11–14. Give extra time to finish the full task set; use Section 15 when the week includes a coding lab.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building. Repeated copies of a case share one recorded attempt; they are not new evidence.

Case exposure check: this source example appears in ai-3/19, ai-3/33. Reuse a familiar case for recall or

modelling; verify that the case used to assess transfer has not already received feedback.

Retrieval answer guidance

Human AI Workflow

Without opening the old lesson, explain one key idea from Human AI Workflow. Give an example and a mistake that the idea helps you avoid.

Look for: Explain the mechanism and terminology of human review and consequential use

Reference example, not the only acceptable wording: Teacher explanation: Human review needs a responsible actor, evidence packet and explicit authority. Adding a person to a workflow does not guarantee effective oversight. For this fictional workflow the assistant drafts a room request but only the coordinator can approve and submit it. Review packets include sources, uncertainty, requested action and the exact version. Any material change requires new approval.

Bias and Representation

Without opening the old lesson, explain one key idea from Bias and Representation. Give an example and a mistake that the idea helps you avoid.

Look for: Explain the mechanism and terminology of representation and bias evaluation

Reference example, not the only acceptable wording: Teacher explanation: Bias evaluation tests defined outcomes across relevant conditions. Group averages can expose differences but cannot alone establish their cause or the correct remedy. Define the task, relevant groups, labels and denominators before comparing. Use fictional, non-identifying records. Evaluate quality and fallback outcomes separately; preserve sample sizes and unresolved explanations.

Red Team the Product

Without opening the old lesson, explain one key idea from Red Team the Product. Give an example and a mistake that the idea helps you avoid.

Look for: Explain the mechanism and terminology of red team exercises

Reference example, not the only acceptable wording: Teacher explanation: Red-team exercises probe explicit failure boundaries in an authorised isolated fixture environment. Their purpose is evidence-based improvement. Use fictional records and allowed test routes. Define a target contract violation, record the exact case, observe action/data traces and preserve a reproducible report. Do not touch external systems.

Use the student module and workbook section with this exact week title. The worked case, independent question and answers below form one aligned lesson sequence.

What the learner must demonstrate

Design a chapter assistant with one answerable, one ambiguous and one unanswerable query. State the expected behaviour and one mandatory quality criterion.

Concept to explain

Requirements, success measures, sources, risks, and tests precede building.

Term · Meaning

Term: requirement

Meaning: A behaviour or condition the proposed system must satisfy.

Term: success measure

Meaning: An observable quantity or criterion used to decide whether a task was achieved.

Term: risk register

Meaning: A record of identified risks, their consequences, mitigations and responsible reviewers.

Term: design review

Meaning: An examination of a proposed system against its requirements before proceeding.

Prepare the class

Use the supplied scenario, student illustrations and blank paper. For numeric work, offer counters or a calculator after learners show the setup. Fictional records are supplied; personal data and paid accounts are unnecessary.

Allow 80 minutes: recall 8; explanation 12; worked demonstration 15; guided practice 20; independent assessment 15; project and reflection 10. Extend the lab into a second session when needed.

Preparation for the connected explanation

Two 45-minute sessions. A grounded-assistant capstone begins with bounded requirements and a versioned permitted corpus. A baseline can be a manual lookup process using the same evidence. Use the supplied constructed scenarios to make the mechanism inspectable. Collect W4 before sharing solutions. Students should name intermediate evidence and limitations; classroom pacing and learning effectiveness remain unvalidated.

Readiness answer

A result depends on the inputs, procedure and conditions that produced it. Preserve data roles and distinguish observed outcomes from predictions or unmeasured conditions. If the prior concept cannot be explained, use a short retrieval activity before this chapter.

Lesson and workbook; paper and pencil

Calculator and optional local Python 3 for arithmetic; no live API or private data

Prepare the fresh retry separately and withhold its solution until after collection

End goal for the pictorial case

By the end, I can explain capstone design using a traced example and a stated limitation.

This added worked case illustrates the week's main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

2. Explain the mechanism

Explain the mechanism

Input → Process → Output

Trace the information. At each stage, ask what changed and how you could test it.

Which stage could fail even when the other two are correct? Explain:

Predict first. Then check each step. A correct answer without visible reasoning is incomplete evidence.

1. Need: answer questions from one approved science unit.
2. Requirements include citations, age-fit explanation, and missing-source abstention.
3. Success: at least 8/10 fixed eval cases pass, with no unsupported claims.
4. Review architecture, privacy, provenance, and safety before building.

Recalculate, retrace, or restate the reasoning in your own representation:

CHECK Did the evidence answer the exact question? Did the total, path, or source record stay consistent?

Concrete teaching case

Capstone: answer from one approved chapter with citations and a missing-evidence fallback.

Define acceptance tests.

Exact explanation to model

Use answerable, ambiguous and unanswerable cases. Check correctness, support, format and fallback, plus declared permissions and privacy.

Explain the added worked example

A grounded-assistant capstone begins with bounded requirements and a versioned permitted corpus. A baseline can be a manual lookup process using the same evidence.

Begin by writing the question the procedure is intended to answer. Identify the information already supplied, the information it will produce and the information still missing. These are constructed classroom examples; they do not require live accounts, private records or an external service. Do not treat a plausible explanation as a measured outcome.

Specify the user question, source authority, access, output/fallback contract and evaluation plan. Freeze corpus IDs and split cases before implementation. Keep teacher material out of student retrieval.

Before applying the procedure, write its units, denominator or decision boundary where relevant. A partner should be able to repeat each step using the same inputs. If a required input is absent or violates the stated contract, record that condition explicitly instead of quietly replacing it with a convenient value.

Key terms

Term · Meaning

Term: Requirement

Meaning: A declared user-visible condition the system must satisfy

Term: Corpus manifest

Meaning: An inventory of source IDs, versions and access roles

Term: Baseline procedure

Meaning: A credible reference process on the same task

Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a prediction, then explain the deciding step. These teacher notes contain solutions; do not reveal the independent case answer during modelling.

Student room assistant corpus contains public register P, teacher-only guide T and current approval A. It must report approved room capacity or needs_review.

Authorised corpus contains P and A, excluding T. Baseline manually checks same-room current approval and capacity. Reserve complete, missing and conflicting cases before development.

Compare the intermediate evidence with the final statement. Explain why each number or decision follows from the stated inputs. The conclusion belongs to this scenario and procedure; changing the inputs, evaluation population or authority can change what it establishes. Retain the raw record so a later revision can be compared honestly.

Thinking move · Teacher prompt

Thinking move: Plan

Teacher prompt: What is given? What is the question? Which procedure fits these facts?

Thinking move: Do

Teacher prompt: Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.

Thinking move: Check

Teacher prompt: Does the output answer the question? Which assumption or limitation prevents a larger claim?

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

A pictorial worked case: Capstone Design

Plan a bounded study buddy around a small approved vocabulary corpus.

The facts we will use

Corpus V1 defines roots, stem and leaf. It contains no plant disease treatment. Success requires a source reference; missing definitions require a clear fallback.

All inputs and outcomes in this worked example are invented classroom data; no real model run is claimed.

Capstone Design

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Define contract

Answer only glossary-definition requests supported by V1.



2. Plan normal test

"What is a stem?" must point to V1's stem entry.



3. Plan missing test

"How do I cure this disease?" must produce an out-of-scope response, not a treatment guess.

CHANGE ONE THING

The corpus gains a teacher-approved flower definition in V2.

Explain the changed result using the matching lesson.

Read the picture one step at a time

Step · What happens and why

Step: 1. Define contract

What happens and why: Answer only glossary-definition requests supported by V1.

Step: 2. Plan normal test

What happens and why: "What is a stem?" must point to V1's stem entry.

Step: 3. Plan missing test

What happens and why: “How do I cure this disease?” must produce an out-of-scope response, not a treatment guess.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Read the picture in order. “How do I cure this disease?” must produce an out-of-scope response, not a treatment guess. The arrows show which recorded input or intermediate result each decision uses. Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

What this example does not establish

A capstone design is a testable plan; its proposed behaviours are not verified outcomes until checked.

Change one thing in the pictured case

Change: The corpus gains a teacher-approved flower definition in V2.

Prediction: Add a flower test using V2 and record the corpus version.

Reason: The evidence boundary changed explicitly; the earlier corpus could not support this new entry.

Ask while showing the picture

Cover the result. Ask learners to name the inputs and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its evidence. Then change the supplied condition and compare. The picture is a representation; it does not document execution of a real AI model.

3. Demonstrate and guide practice

Demonstrate and guide practice

Board sequence

Write the given facts: Capstone: answer from one approved chapter with citations and a missing-evidence fallback.

Write the question to explain: Define acceptance tests.

Reveal after predictions: Use answerable, ambiguous and unanswerable cases. Check correctness, support, format and fallback, plus declared permissions and privacy.

Keep for an independent attempt: The capstone goal is helps students. Rewrite it as an observable test for a passage-based study assistant.

Classroom dialogue

Ask: Which facts are given, and which would we have to assume? Learners mark relevant details before discussing answers.

Say: Predict first. Show the rule, information path or calculation that would produce the result. A correct guess is not yet an explanation.

Reveal reasoning one step at a time. Explain why a tempting alternative fails. Write units and denominators for arithmetic, trace conditions for decisions, and identify supporting sources for claims.

In pairs, one learner solves while the other checks evidence; exchange roles. Remove the worked answer before independent assessment.

Model these guided questions

Guided: Authorised corpus contains P and A, excluding T. Baseline manually checks same-room current approval and capacity. Reserve complete, missing and conflicting cases before development. Qualification: A complete capstone plan is not evidence that the implementation meets it; evaluate after building. Exit: accept a concrete variation with its predicted outcome and a stated condition that would challenge the mechanism; reject a claim resting on unmeasured evidence.

Case-specific teaching sequence

Stage · Teacher action

Stage: Session 1: 0–7 minutes

Teacher action: Retrieve the prerequisite; identify the input, procedure and evidence source.

Stage: 7–18 minutes

Teacher action: Explain section 1 and vocabulary. Complete W1 with a mechanism-based explanation.

Stage: 18–32 minutes

Teacher action: Trace the declared procedure and worked case. Complete W2, recording intermediate steps.

Stage: 32–45 minutes

Teacher action: Discuss the diagram with the worked input; use W3 to distinguish measured claims and limitations.

Stage: Session 2: 0–8 minutes

Teacher action: Retrieve the worked reasoning and collect the guided response before discussing it.

Stage: 8–20 minutes

Teacher action: Compare a proposed boundary variant with the original case; identify what remains fixed.

Stage: 20–35 minutes

Teacher action: Collect W4 independently. Require the procedure trace and the bounded conclusion.

Stage: 35–45 minutes

Teacher action: Use W5 and exit responses for feedback. Administer the fresh retry after feedback and preserve both attempts.

Reduce help deliberately

Completion task: Student room assistant corpus contains public register P, teacher-only guide T and current approval A. It must report approved room capacity or needs_review. Show the intermediate steps and give the conclusion with its scope.

Teacher check: Teacher explanation: Authorised corpus contains P and A, excluding T. Baseline manually checks same-room current approval and capacity. Reserve complete, missing and conflicting cases before development.

Start by identifying the given inputs together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood. For partners, alternate explainer and checker; collect a response from every learner before discussion.

Individual reasoning check

Use the worked case for Capstone requirements and corpus. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

Expected correction: A complete capstone plan is not evidence that the implementation meets it; evaluate after building.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class. If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

Pictorial practice question

What should the project plan record before the new flower test? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.

Reasoned teacher answer

Record approved corpus V2, the expected sourced definition and which evaluation cases belong to that version. Changed-case check: Add a flower test using V2 and record the corpus version. The evidence boundary changed explicitly; the earlier corpus could not support this new entry.

Changed-case reasoning: Add a flower test using V2 and record the corpus version. The evidence boundary changed explicitly; the earlier corpus could not support this new entry.

Collect each learner's explanation before discussion. Use the first missing connection between input, deciding step and result to choose support. This question is worked-case practice, not fresh mastery evidence.

4. Run the weekly evidence lab

Run the weekly evidence lab

Words You Can Use to Reason

A definition matters when you can recognize the idea, use it, and contrast it with a near-miss.

Predict → Test → Record → Explain

Keep expected and actual results separate. Surprises are useful data when recorded honestly.

Write the brief, architecture, source list, schema, rubric, eval set, risks, and approval gates; peer-review for testability.

What did the hardest test teach you that the normal test did not?

Make the lab runnable

Use the concrete case as the shared normal case and the independent question as a second case. Each pair creates a boundary or missing-information case and predicts its result before testing. Keep the expected result separate from an observed mistake.

Record · What to capture

Record: Input or evidence

What to capture: Exact data, instruction or claim

Record: Expected result

What to capture: Answer or safe fallback declared before the run

Record: Observed result

What to capture: Actual result, including errors

Record: Explanation

What to capture: Rule, calculation or evidence explaining the outcome

Record: Next test

What to capture: One justified change and a new prediction

Diagnose and reteach

For guessing, return to given facts and request a trace. For vocabulary without application, use one example and one non-example. For arithmetic errors, check the setup before the calculation. Finish with a different uncoached case before recording mastery.

Address these additional misconceptions

The worked result establishes every future case: A complete capstone plan is not evidence that the implementation meets it; evaluate after building.

A missing input may be replaced without recording the change: A replacement changes the evidence or contract. Record the change and its justification; do not silently turn unknown into a measured value.

Repeating an assessed case is new independent evidence: A replay checks consistency. An example used for feedback or revision cannot later be described as untouched final evaluation.

Match the support to the first error

What the response shows · Next teaching action

What the response shows: Missing or misread facts

Next teaching action: Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.

What the response shows: Wrong procedure

Next teaching action: Return to the deciding step in the worked case. Contrast it with the proposed

incorrect explanation.

What the response shows: Arithmetic or implementation error

Next teaching action: Trace intermediate values or lines; preserve the first mismatch and retry that step.

What the response shows: Correct result without a reason

Next teaching action: Ask for a trace and a changed case before marking independent understanding.

What the response shows: An unsupported wider claim

Next teaching action: Ask which supplied evidence supports the claim and which further observation would be needed.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal. Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

5. Assess and explain the answers

Assess and explain the answers

Worked case answer

Use answerable, ambiguous and unanswerable cases. Check correctness, support, format and fallback, plus declared permissions and privacy.

Independent question

The capstone goal is helps students. Rewrite it as an observable test for a passage-based study assistant.

Expected independent answer

For example: given five questions answerable from a supplied passage, answer each with supporting passage evidence, and abstain on two questions the passage cannot answer. State expected outputs before testing.

What counts as evidence

Criterion · 0 points · 1 point · 2 points

Criterion: Result

0 points: Incorrect or absent

1 point: Partly correct

2 points: Correct with necessary conditions

Criterion: Reasoning

0 points: No supporting trace

1 point: Some steps

2 points: Clear mechanism, calculation or source support

Criterion: Limit

0 points: Overclaims certainty

1 point: Vague limitation

2 points: Relevant uncertainty or missing fact

Criterion: Transfer

0 points: Copies without adapting

1 point: Adapts with prompts

2 points: Solves a changed case independently

Use 6/8 as a classroom checkpoint, with 2 required for both Result and Reasoning. This is a proposed teaching rule, not a validated certification threshold. Do not count a coached answer as independent mastery; correct unsafe or unsupported claims regardless of total.

Feedback

Ask the learner to find the evidence that challenges this claim: “A broad goal is already a measurable acceptance criterion.” Keep the first attempt and an uncoached retry separate.

Original transfer answer

Supply a direct fact query with a supporting passage; an unclear 'it' query requiring clarification; and an absent fact requiring fallback. Require every factual answer claim to be supported by its cited passage. Preserve the cases before implementation.

Explanations for the additional workbook examples

W1 Explain the mechanism

Explain the weekly mechanism in your own words. Use the worked scenario to name its input, decision rule and output.

Teacher explanation: A grounded-assistant capstone begins with bounded requirements and a versioned permitted corpus. A baseline can be a manual lookup process using the same evidence. Specify the user question, source authority, access, output/fallback contract and evaluation plan. Freeze corpus IDs and split cases before implementation. Keep teacher material out of student retrieval.

W2 Trace the worked case

Student room assistant corpus contains public register P, teacher-only guide T and current approval A. It must report approved room capacity or needs_review. Show the intermediate steps and give the conclusion with its scope.

Teacher explanation: Authorised corpus contains P and A, excluding T. Baseline manually checks same-room current approval and capacity. Reserve complete, missing and conflicting cases before development.

W3 Audit the limit

A report omits this qualification: A complete capstone plan is not evidence that the implementation meets it; evaluate after building. Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

Teacher explanation: A complete capstone plan is not evidence that the implementation meets it; evaluate after building. State the observed result separately from the unverified or out-of-scope claim; preserve the original evidence rather than inventing a missing fact.

W4 Apply a new case independently

Student equipment assistant has public inventory I, restricted purchase ledger L and public booking notice B. Define eligible corpus, baseline and a missing-evidence test.

Teacher explanation: Use I and B only; exclude L. Baseline manually matches item availability to authorised booking evidence. Missing notice should trigger the specified review outcome.

W5 Design a revealing follow-up

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison. Do not reuse W4 as a new final test after receiving feedback.

Teacher explanation: Accept a concrete variant of the worked input or a missing/conflicting/boundary condition, with an expectation derived from the declared procedure. Require preserved settings, original outputs and data roles. A complete capstone plan is not evidence that the implementation meets it; evaluate after building. The new case should test this particular mechanism rather than merely restate the

original result.

Assess independent transfer

Selected case: A science assistant may use only these notes: evaporation is liquid becoming vapour; condensation is vapour becoming liquid. Specify behavior for “Define evaporation”, “Explain it”, and “What is tomorrow’s weather?”.

Selected case answer

Answer the definition using the supplied note and identify its source. Clarify the referent of “it”. Weather is absent from these notes: say the evidence is unavailable rather than inventing a forecast. Require support for every factual claim.

Dimension · Evidence required

Dimension: Explanation

Evidence required: Names the applicable idea or procedure and ties it to the supplied facts.

Dimension: Trace and result

Evidence required: Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.

Dimension: Limits

Evidence required: States the assumptions, missing evidence or conditions under which the result would change.

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace. Do not penalise an honestly recorded model failure if the learner correctly explains its evidence and limits.

Feedback that the learner uses

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case. Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

6. Connect and extend learning

Connect and extend learning

Verified Study Buddy

Freeze the Study Buddy v1 design and acceptance tests before implementation.

Keep the decision, input, result, explanation and remaining limitation in the project evidence record.

Retrieve this record next week rather than restarting the project.

Support and stretch

Provide one short approved passage; separate instruction, input, output and evidence. Stretch: introduce an unsupported claim or conflicting source and validate claim by claim with a fallback.

Exit reflection

Ask for one decision the learner can now justify and one situation requiring more evidence or human review. Confidence ratings are separate from correct independent performance.

Bridge to the next week

Next: Week 34: Capstone Build. Retrieve today's evidence and identify what carries forward and what is new.

Support and fresh reassessment

Provide an input/procedure/result/limit frame and a calculator when needed. Ask students to identify the first step before completing the whole trace. Extension: compare two concrete variants of this mechanism while holding unrelated conditions fixed; report both successful and unsuccessful outcomes.

W4: Use I and B only; exclude L. Baseline manually matches item availability to authorised booking evidence. Missing notice should trigger the specified review outcome. Require intermediate reasoning and the stated scope. Fresh retry: Public timetable S and restricted staff notes N; student assistant reports meeting time. Expected reasoning: Use S only. Manual timetable lookup is baseline; missing current timetable gives fallback rather than staff-note access. If students need support, record it and reassess with another new case rather than awarding independence to a copied answer.

Next connection: Capstone grounded assistant. Preserve the current case record and retrieve its evidence boundary before moving on.

Plan the delayed return

Suggested return points in this level: ai-3/34, ai-3/36. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used. For a new level, revisit the previous capstone before checking the new prerequisites.

Record · Teacher evidence

Record: Learner code and date

Teacher evidence: _____

Record: Concept and case used; prior exposure

Teacher evidence: _____

Record: Explanation / trace / limits

Teacher evidence: _____

Record: Support and first repair

Teacher evidence: _____

Record: Changed case and delayed outcome

Teacher evidence: _____

Record: Next teaching action

Teacher evidence: _____

Use the evidence to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

Project evidence from this case

For your Verified Study Buddy, save the input, procedure, observed result and limitation of this check. Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.